MANTHAN DUBEY

Software Engineer · Full-Stack Systems & Real-Time Architectures

📍 Bengaluru / Lucknow, India ✉️ manthan1181999@gmail.com 📞 +91 6393593832 🔗 linkedin.com/in/dubeymnthn

💻 github.com/dubeymnthn 🌐 builtbymanthan.netlify.app

PROFESSIONAL SUMMARY

Full-stack software engineer with a strong track record of designing, building, and operating production web applications, real-time distributed microservices, and enterprise middleware. Experienced across healthcare data platforms, core banking integrations (ANZ Bank via IBM), and B2B wholesale systems. Proficient in engineering end-to-end architectures utilizing TypeScript, React, Next.js, Node.js, Fastify, Flutter, PostgreSQL, and Docker. Passionate about clean domain boundaries, concurrency safety, low-latency communication, and high-quality software craftsmanship.

TECHNICAL EXPERTISE

LANGUAGES	TypeScript, JavaScript (ES6+), Dart, Python, C++, SQL, HTML5, CSS3/SCSS
FRONTEND	React, Next.js (App Router), Redux Toolkit, Vite, Tailwind CSS, Ant Design, Web Audio API
BACKEND	Node.js, Fastify, Express, RESTful APIs, WebSockets (Socket.IO), Redis Pub/Sub, BullMQ
MOBILE & SYSTEMS	Flutter, Android SDK, Android NDK / JNI (C++ Game Engine), Progressive Web Apps (PWA)
DATABASES & ORM	PostgreSQL, Prisma ORM, MongoDB, SQLite / SQLCipher, MySQL, Database Transactions
CLOUD & DEVOPS	Docker, Docker Compose, AWS (EC2, S3, Cognito), CI/CD Pipelines, Linux Shell, IBM DataPower, IBM UrbanCode Deploy (UDeploy)
AI & PROTOCOLS	Model Context Protocol (MCP), LLM Integration (Google Gemini, Ollama), Prompt Engineering, Headless Automation (Selenium, PDFPlumber)

PROFESSIONAL EXPERIENCE

IHX — A Perfios Company · Bengaluru, India
March 2026 – Present

Software Engineer (Full-Stack Developer)

- **Healthcare Claims Platform:** Engineered full-stack features for healthcare claims processing and verification portals connecting medical providers, insurers, and clinical backends.
- **Microservices & Scalability:** Architected and supported Node.js backend microservices handling structured insurance document payloads and real-time validation pipelines.
- **Frontend Applications:** Built responsive, high-performance web applications using React and modern component libraries, improving clinical workflow efficiency.
- **Data Integrity:** Standardized API schema validation layers to isolate malformed payloads prior to persistence, ensuring enterprise-grade auditability and reliability.

IBM / ObjectWin Technologies (Client: ANZ Bank) · Bengaluru, India
June 2025 – March 2026

Contract Software Developer (Released March 13)

- **Banking Middleware:** Developed and supported Node.js-based middleware services (MU2) supporting high-security institutional payment workflows and settlement systems for ANZ Bank.
- **Enterprise Onboarding:** Integrated backend banking microservices to onboard corporate clients onto the centralized payments platform.
- **API Gateway Security:** Configured API routing, payload transformations, and transport security policies using IBM DataPower gateways.
- **Release Engineering:** Conducted scheduled production deployments and release activities using IBM UrbanCode Deploy (UDeploy) and Connect:Direct under rigorous change management governance.

Karkinos Healthcare Pvt. Ltd. · Kochi / Bengaluru, India
Dec 2022 – Jan 2025

Associate Software Developer (2 yrs 2 mos)

- **Oncology Platform Engineering:** Developed and maintained clinical web applications using React on the frontend and Node.js/Express for backend microservices.
- **State Management:** Implemented centralized Redux state architecture across complex multi-step oncology consultation workflows, eliminating rendering bottlenecks and state synchronization issues.
- **Clinical PWA & Biometrics:** Built a Progressive Web Application (PWA) with offline caching for field staff and integrated national ABHA biometric authentication APIs.
- **Database Optimization & CI/CD:** Structured RESTful APIs and MongoDB collections; optimized caching and set up containerized Docker/AWS pipelines for staging and production.

MANTHAN DUBEY · SOFTWARE ENGINEER

✉ manthan1181999@gmail.com 📞 +91 6393593832

🔗 builtbymanthan.netlify.app

SELECTED COMPLETED ENGINEERING PROJECTS

01. Wally Talky — Push-to-Talk Real-Time Audio Platform

`Flutter · Fastify · Socket.IO · PostgreSQL · Redis · FCM`

Architecture: Event-driven voice communication platform pairing a reactive Flutter mobile client with a Fastify/TypeScript gateway. Control events and audio packets flow through WebSocket channels backed by Redis Pub/Sub for sub-second room fanout.

- Engineered low-latency push-to-talk voice capture workflow with asynchronous audio stream buffering in Flutter.
- Constructed real-time user presence and heartbeat monitoring using Socket.IO and Redis pub/sub.
- Built chunked audio storage pipeline generating HMAC-signed temporary playback URLs for access control.
- Integrated high-priority Firebase Cloud Messaging (FCM) background wakeup handlers for incoming mobile transmissions.

02. StoryForge AI — Hindi Audio Drama Production Studio

`React 19 · Express · Google Gemini · Web Audio API · SSML`

Architecture: Multi-stage creative engineering pipeline converting story premises into structured character bibles, multi-act Hindi scripts, custom TTS voice profiles, and multi-track audio timelines.

- Engineered task-specific ModelRouter coordinating Google Gemini models with strict JSON schema serialization.
- Created the AntiSlopDirector module auditing dialogue against conversational Hindi idioms and eliminating synthetic clichés.
- Developed client-side multi-track Web Audio API timeline for auditioning, splicing, and mastering voice takes with background ambience.

03. Dravyam (GodownOS) — B2B Wholesale Inventory Platform

`Fastify · PostgreSQL · Prisma (35 Models) · BullMQ · React 19 · Next.js`

Architecture: Enterprise B2B wholesale platform managing multi-warehouse godown operations with granular visibility controls preventing wholesale price discovery by competitor retailers.

- Architected 35 Prisma relational models with atomic double-entry inventory ledger logic (Available = On Hand - Reserved).
- Enforced backend visibility and price tier matrices dynamically inside Prisma query filters, eliminating client-side data leaks.
- Constructed asynchronous BullMQ worker queues for inventory reconciliation, batch CSV imports, and background jobs.

04. Financial Data Automation & Statement Reconciliation

`Python · Selenium · PDFPlumber · Pandas · MongoDB · Node.js`

Architecture: High-resilience data extraction pipeline harvesting and parsing broker contract notes, trade ledgers, and F&O statements from AngelOne, IIFL, and banking portals.

- Constructed resilient Selenium browser automation handling dynamic 2FA checkpoints, session renewal, and downloads.
- Engineered adaptive X-coordinate density clustering in PDFPlumber to accurately isolate tables across non-bordered PDF pages.
- Built F&O profit/loss reconciliation calculating realized/unrealized gains, STT fees, and GST deductions into MongoDB.

05. MCP Token Server & Chrome Extension

`Node.js · Express · Chrome MV3 · Model Context Protocol · TypeScript`

Architecture: Localhost authentication bridge daemon connecting browser session authentication with local developer tools and AI coding agents via standardized Model Context Protocol (MCP) tool endpoints.

- Engineered Manifest V3 extension capturing authorized session cookies; transmitted securely over localhost loopback socket.
- Stored credentials exclusively in ephemeral RAM with automated TTL expiry, eliminating plain-text disk exposure.
- Exposed standard MCP tools (get_auth_token, verify_session) allowing IDE assistants to execute authenticated workflows.

EDUCATION

Master of Computer Applications (MCA) · Vellore Institute of Technology (VIT) 2021 – 2023

Bachelor of Computer Applications (BCA) · National Post Graduate College, Lucknow University 2018 – 2021

EARLY ENTREPRENEURSHIP

Sandhara · Lucknow, India January 2020 – September 2021

Founder

Established an agricultural supply chain initiative during undergraduate studies, directly connecting regional farmers' markets with urban consumers and retailers. Built B2B2C distribution logistics, managed multi-tier pricing, and coordinated regional fulfillment.